

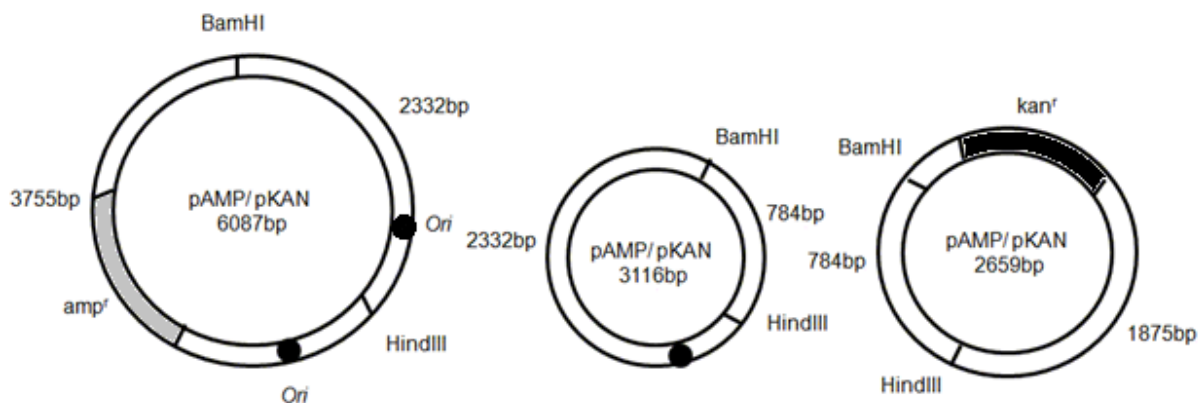


**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2016
YEAR SIX
H2 BIOLOGY (9648)
PAPER 3**

Structured Questions Answers

Question 1

(a)(i)



1M for each drawing of recombinant plasmid showing size of plasmid, correct labels, position of antibiotic resistance gene and *ori*

Comment: many students draw without showing complete labeling. The relative size of plasmids and segments should also be accurately represented.

(ii)

Culture *E. coli* on a nutrient plate containing ampicillin and kanamycin;

Plasmid shown in **Fig 1.2** contains ampicillin and kanamycin resistant genes and hence *E. coli* carrying this type of plasmid will survive and grow into colonies;

Or

E. coli carrying the other three types of plasmid does not contain both ampicillin and kanamycin resistant genes) and thus will not survive and grow into colonies;

Note: Plasmids DO NOT survive and grow on nutrient plates.

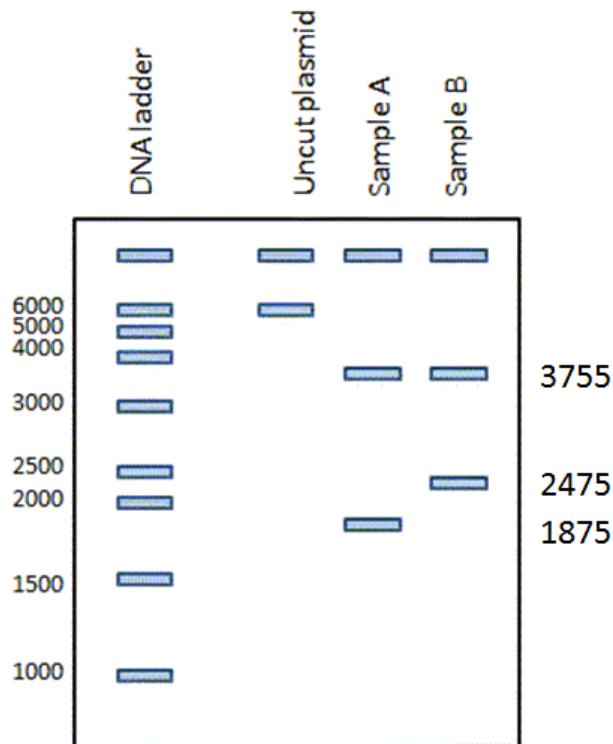
(iii)

Sample B is able to survive in an agar plate containing ampicillin but not kanamycin;

Sample B bacteria carry plasmid with human gene. 3755 kb and 2475 kb fragments. This 2475 kb fragment is the 1875 kb with an addition of the 600kb human gene;

Thus *kan^r* gene is disrupted by the insertion of the human gene and that the bacteria cannot product protein that confer resistance to kanamycin and thus not able to grow on the plate with kanamycin;

(iv)



(b)(i)

Reverse transcriptase – synthesizes single stranded cDNA from mRNA template;

RNAse- partially degrades mRNA template after single stranded cDNA is synthesized, remaining short RNA segments serve as primers for synthesis of second strand;

DNA polymerase – synthesizes second strand of cDNA using first strand of cDNA as a template through the formation of phosphodiester bond;

DNA ligase – joins cDNA fragments of second strand to get a complete double-stranded cDNA molecule through the formation of phosphodiester bond;

1 Max.

(ii)

The starting material / template for the synthesis of cDNA is mature mRNA expressed from certain genes at that particular time;

The expression of genes varies at different times in the life of the cell / can also change in response to different stimuli;

For genomic library, the set of DNA remains the same throughout the life of the cell;

Question 2

(a)(i)

Human beings are diploid organisms where 2 different alleles are inherited, one from each parent;

(ii)

Centromere, telomere, introns;

1 Max.

(b)(i)

Excess primers are used where peak 1 corresponds to the primers;

And peak 2 corresponds to the primer dimers;

(ii)

For PCR products corresponding to each peak, there is a need to consider the length of the 2 flanking primers, i.e.

$15 \text{ bp} \times 2 = 30 \text{ bp}$;

Since primers are not repeats, there is a need to subtract the length of the flanking primers from the total length of the PCR products, i.e.

For peak 3: $443 \text{ bp} - 30 \text{ bp} = 413 \text{ bp}$

For peak 4: $590 \text{ bp} - 30 \text{ bp} = 560 \text{ bp}$;

Since each repeat is 16 bp in length, divide length of PCR products in step 2 by 16 to obtain number of repeats present in each VNTR allele:

Number of repeats present in VNTR allele in peak 3 = $413/16 = 25.8$ (correct to 3 s.f.) = 26

Number of repeats present in VNTR allele in peak 4 = $560/16 = 35$;

(iii)

Identify as the 443 bp fragment where it has 25.8 repeats, i.e. not whole repeats;

This could be due to deletion mutation;

(iv)

Small amounts of DNA collected from volunteers, therefore need amplification for differences to be detected;

No restriction sites flanking the locus, thus restriction enzymes cannot cut out the marker / locus for RFLP analysis;

Specific primers used, thus can home in on a single marker / locus;

No need for Southern blot, thus it is less tedious / safer because radioactive probes are not used;

3 Max.

Question 3

(a)(i)

- Deficiency of ADA results in accumulation of the substrate deoxyadenosine (dA) in cells. This leads to the buildup of dATP in cells;
- dATP inhibits ribonucleotide reductase and prevents DNA synthesis / formation of deoxyribonucleoside diphosphates (dNDP);
- As a result, T lymphocytes are unable to divide. Consequently, the immune system is severely compromised and this leads to SCID;

(a)(ii)

- Use hematopoietic stem cells from the bone marrow of the patient instead of T-lymphocytes;
R! Bone marrow cells / bone marrow stem cells
- since hematopoietic stem cells are able to self renew and differentiate into B- and T-lymphocytes. This will provide a more long term / permanent cure;

(b)(i)

When mutant allele is recessive, addition of functional / normal dominant allele by gene therapy will produce sufficient amounts of gene product to mask the effect of recessive allele;

If mutant allele is dominant, both alleles must be removed / repaired / inactivated to block production of defective gene product;

(b)(ii)

- Transient expression of therapeutic gene / difficult to ensure that therapeutic gene is integrated into genome of host cells;
- Incorrect insertion of therapeutic gene into genome of host cell results in cancer / insertional mutagenesis, as the insertion is random;
- Immune response may be triggered resulting in rejection because a foreign vector is introduced;
- Difficult to control expression of normal functional gene to give a fully functional protein;
- Unable to treat multigene disorders;

3 Max.

(c)(i)

- iPS cells can be generated from skin cells/adult cells, so there are **fewer ethical issues** involved as compared to using embryonic stem cells obtained from an embryo;
- iPS cells can be generated from skin cells/adult cells of the patient/sufferer, so there might be **less risk of tissue rejection** after gene therapy and transplantation;
- iPS cells are more **readily available** than embryonic stem cells; OWTTE

(c)(ii)

1. Each transcription factor can **activate the transcription of multiple genes**. Each gene in turn could code for a transcription factor which **activates/inactivate other genes**;
2. Transcription factors **switch on genes that are expressed in ES cells**, resulting in the synthesis of proteins which are found in ES cell;
3. **Give example of activated genes:** e.g. telomerase gene/ genes which promote cell division/ genes which cause the cell to revert to undifferentiated state;
4. Transcription factors could also **inhibit gene expression / repress genes not expressed in ES cells**;
5. **Give example of inactivated genes:** e.g. genes that result in differentiation / specialisation;

3 Max.

Planning Answer

Theory	The hydrophobic core of the phospholipid bilayer of the cell membrane prevents any large, polar, hydrophilic molecules like betacyanin to freely exit the plant cell. Cell membrane follows fluid mosaic model - composed of membrane proteins embedded in a phospholipid bilayer. Increase in temperature increases kinetic energy, resulting in increased movement of phospholipids, increasing fluidity of membrane, making membrane slightly leaky. Too high a temperature can cause movement of phospholipids to be too great, disrupting membrane integrity, causing betacyanin to leak out of the cell. Too high a temperature can also cause denaturation of membrane proteins by disrupting hydrophobic interactions, hydrogen bonds and ionic bonds, also disrupting membrane integrity, causing betacyanin to leak out of the cell. As temperature increases, the cell surface membrane will become more fluid and leak a small amount of pigment, beyond a certain temperature the membrane integrity would become significantly disrupted and a large amount of pigment would leak out of the cell into the bathing solution. Beyond a certain temperature, membrane integrity is completely disrupted and further increase in temperature will not result in significant increase in amount of betacyanin released.	1. Theory on why betacyanin cannot pass through membrane 2. theory of how temperature affects membrane phospholipids OR theory of how temperature denatures membrane proteins 3. hypothesis:																												
Variables	Independent variables: 5 temperature water baths, evenly distributed (10, 30, 50, 70, 90°C). Dependent variables: intensity of red colouration of bathing solution after a fixed time interval in waterbath (as measured using colorimeter / as compared against colour standard) Controlled variables: Fixed amount of time given for soaking in bathing solution, Number/size of discs of beetroot used,	4. independent and dependent variables																												
Procedure	1. Use the cork borer to obtain a uniform cylinder of beetroot. 2. Use a ruler to measure 1cm to ensure uniform thickness and a scalpel to cut the beetroot into discs. Obtain 3x 5 discs. 3. Wash the discs and ensure that any red pigment leakage caused by cutting is washed away before start of experiment. 4. Carryout dilution of the 5% betacyanin to have a colour standard for comparison according to the table below. <table><tr><td>Percentage of betacyanin / %</td><td>Volume of 5% betacyanin stock / cm³</td><td>Volume of distilled water / cm³</td><td>Final volume / cm³</td></tr><tr><td>5</td><td>10</td><td>0</td><td>10</td></tr><tr><td>4</td><td>8</td><td>2</td><td>10</td></tr><tr><td>3</td><td>6</td><td>4</td><td>10</td></tr><tr><td>2</td><td>4</td><td>6</td><td>10</td></tr><tr><td>1</td><td>2</td><td>8</td><td>10</td></tr><tr><td>0</td><td>0</td><td>10</td><td>10</td></tr></table>	Percentage of betacyanin / %	Volume of 5% betacyanin stock / cm ³	Volume of distilled water / cm ³	Final volume / cm ³	5	10	0	10	4	8	2	10	3	6	4	10	2	4	6	10	1	2	8	10	0	0	10	10	5. how to ensure uniform size of discs 6. wash 7. dilution of betacyanin for color standard
Percentage of betacyanin / %	Volume of 5% betacyanin stock / cm ³	Volume of distilled water / cm ³	Final volume / cm ³																											
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2	4	6	10																											
1	2	8	10																											
0	0	10	10																											

	5. Label 5 large beakers (10, 30, 50, 70, 90°C) and prepare the 5 different water bath temperatures by mixing hot water and tap water or ice provided. Use a thermometer to check and monitor water bath temperatures. Add ice or hot water as necessary to maintain these temperatures. OR Prepare the 5 different thermostatically controlled water baths at 10, 30, 50, 70, 90°C. Use a thermometer to check and adjust settings to achieve and maintain water bath temperatures. 6. Label 3x 5 test tubes (10, 30, 50, 70, 90°C) and add 3cm³ of distilled water into each tube using a clean syringe. 7. Place the test tubes in each of their respective water baths and incubate for 5 minutes to allow distilled water in each test tube to reach water bath temperature. Use thermometer to check that test tube distilled water has reached desired temperature. The diagram shows a cross-section of a rectangular container representing a water bath. Inside, a test tube is partially submerged. A thermometer is also in the bath, with its bulb in the liquid. A small dark circle representing a beetroot disc is at the bottom of the test tube, surrounded by liquid. Labels with leader lines point to the 'Thermometer', 'Test tube', 'Beetroot disc in 3ml of distilled water', and the container itself, which is labeled 'Thermostatically controlled waterbath OR Waterbath in beaker'. 8. Use the forceps to transfer a beetroot disc into first test tube at 10°C. Immediately start the stop watch and allow the discs to incubate for 5 minutes. 9. After 5 minutes of incubation time, remove beetroot disc from test tube using forceps / decant bathing solution from test tube into clean test tube. 10. Observing the test tube against a white card as background, compare with prepared colour standard to determine concentration of betacyanin present. OR fill a cuvette with 1ml (A:1ml-1.5ml) distilled water and place in colorimeter. Press tare button. Fill a second cuvette with 1ml (A:1ml-1.5ml) of beetroot bathing solution. Place the cuvette in the colorimeter, press the test button and take the absorbance reading. 11. Record results in the table below. 12. Repeat steps 8-9 to obtain 3 replicates for each temperature to ensure reliability of results. Repeat experiment 2 more times to ensure reproducibility of results. 13. Plot a graph as shown below.	8. how to prepare & maintain water baths 9. equilibrate temperate 10. relevant figure 11. how to ensure uniform incubation time 12a. comparison with colour standard OR 12b1. tare colorimeter with water 12b2. 1ml in cuvette to take absorbance reading w/o beetroot. 13. 3 replicates, 2 repeats
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Table showing effect of temperature / °C on percentage of betacyanin found in bathing solution /%

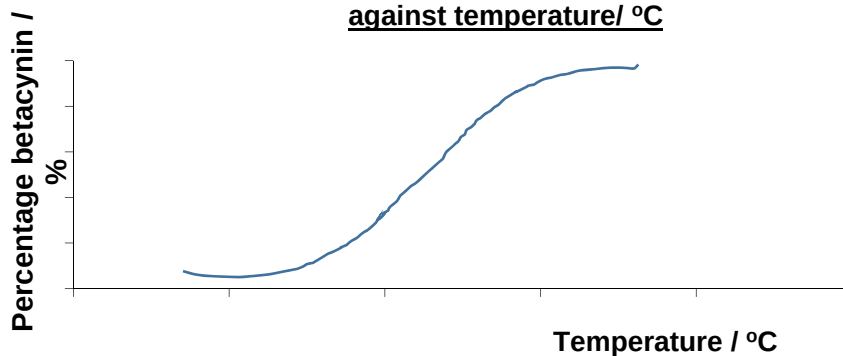
Temperature / °C	percentage of betacyanin / %			
	Reading 1	Reading 2	Reading 3	Average
10				
30				
50				
70				
90				

OR

Table showing absorbance readings of bathing solutions after 5min incubation at varying temperatures / °C

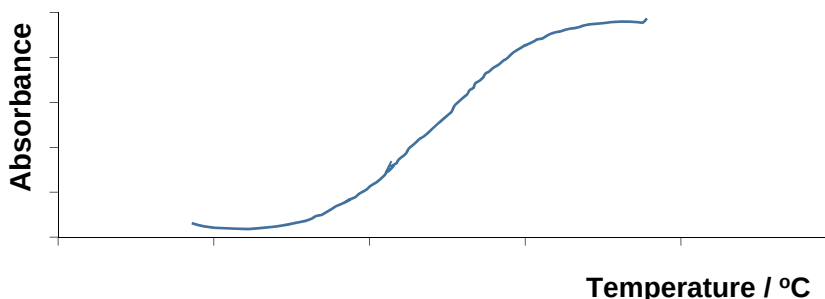
Temperature / °C	Absorbance			
	Reading 1	Reading 2	Reading 3	Average
10				
30				
50				
70				
90				

Graph of percentage betacyanin in bathing solution/% against temperature/ °C



OR

Graph of absorbance against temperature / °C



14. appropriate table with correct headings with units for recording observation of red colouration

15. graph with correct headings / units

16. Predicted trend

Risk and precaution

Cut beetroot using scalpel against a white tile in a direction away from your body (side-ways or downwards) and others to prevent injury.

Use oven mittens when handling beaker with hot water to prevent getting scalded.

17. Safety

Essay Answers

5(a) Biofuels can be obtained from the conversion of cellulosic biomass, which is both abundant and renewable. However, the enzymes and pretreatment processes involved are very expensive. One approach is to genetically engineer plants to produce cellulase so as to enhance the conversion of cellulose into fermentable sugars and reduce the need for pretreatment processes.

Describe how one can mass produce more plants that has already been genetically modified to synthesize cellulase. [7]

1. Obtain an **explant from plant meristematic tissue** (e.g. shoot tip) from plant that has already been genetically modified to synthesize cellulase;
2. **Culture in a medium** (with mineral nutrients, carbohydrate source, and plant growth regulators);
3. under **aseptic conditions** (e.g. surface sterilization, use of laminar flow cabinet);
4. Formation of **mass of undifferentiated cells** known as **callus**;
5. **Subculture** to increase number of callus with desired gene;
6. **Increase cytokinin to auxin ratio for shoot formation**;
7. **Increase auxin to cytokinin ratio for root formation**;
8. **Acclimatize plantlets** by growing in **sterile soil** in a greenhouse;

7 Max.

5(b) Explain the significance of genetic engineering in GM salmon and Bt corn in solving the demand for food in the world. [7]

Inability to cope with demand for food in the world

1. Food production must increase in order to cope with the **increase in human population** as **traditional methods for growing food may not be sufficient** to meet the demands;

GM salmon

2. In normal salmon, the gene that controls the production of growth hormone is activated by light, so the **fish usually grow only during the warm summer months**;
3. Genetically engineer GM salmon by the insertion of a **growth hormone gene** from a Pacific Chinook salmon and an **active promoter** from an ocean pout placed upstream of the growth hormone gene;
4. The active promoter allows the **growth hormone to be expressed all year round / GM salmon can grow all year round**;
5. This results in the GM salmon **reaching market size in a shorter time**, thereby **increasing the supply of salmon**;

Bt corn

6. **Insects can cause damage to crops** both in the field and during storage in silos;
7. Genetically engineer **maize/potato** by insertion of a **Bt-toxin gene** from the bacteria ***Bacillus thuringiensis*** that codes for the production of a crystalline Bt-toxin protein;
8. When insects eat Bt toxins, **toxin is broken down by the digestive enzymes into toxic proteins** that paralyzes the insect's digestive system and forms holes in the gut wall, **killing the insect**;
9. This results in **increased crop yield and quality**;

7 Max.

5(c) Discuss the social and ethical implications of Bt corn. [6]

1. **Bt toxic effects on non-target organisms;**

- Organisms that are predators and parasites of pests are of benefit to agriculture, helping to regulate the population of the pests. However, unforeseen effects of the accumulation of Bt toxins on these organisms could cause them to die instead.

2. **Religious / Dietary restrictions** in food choices;

- Some religious and ethnic groups have restrictions in the food that they can consume due to their religious/personal beliefs. GM food may further complicate their food choices.

3. **Allergies** to new proteins synthesised;

- The Bt protein may lead to unexpected allergic reactions that consumers may not be aware of.

4. **Monopolization / Concentration of economic power** into a few large multinational companies;

- As GM research is heavily funded by private companies, there is a fear that there might be **monopolization of agriculture by certain companies** since research and production of GM food is impeded by the protection of intellectual property.
- There may also be possible **conflicts of interest** between the need for a private company to make money and the application of privately owned technology to solving food and economic problems in poor countries.

5. **Erosion of rural communities;**

- Increased usage of GM food may lead to increased dependence on industrialized nations by developing countries. Thus, Bt corn can be as a power grab that threatens the sustainability of rural communities.
- Alternatively, increased cultivation of Bt corn may lead to increased deforestation for farming and transport infrastructure, forcing smaller farmers out of their jobs.
- Cultivation of other crops may also decrease, increasing the income divide due to the reliance on foreign imports.

6. **Tampering with nature / “Playing God”;**

- Some religious groups have strong moral objections to scientists moving DNA from one species to another, breaking the natural species barrier.

7. **Labeling of GM food;**

- Some members of the public may be fearful of eating ‘weird’ food and refuse to eat Bt corn. These people would like GM food to be labeled for transparency.
- However, proponents of GM foods feel that there is “no difference” between GM crops and traditionally-bred crops and thus there is no need for labeling.

AVP (1 mark per implication with elaboration);

6 Max.